XIAOQI JIANG

pronounced *shyow-chee jyahng*

Data Science Institute, The University of Chicago
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EDUCATION

The University of Chicago Expected Dec. 2026
M.S. in Data Science, Data Science Institute — GPA: 4.0/4.0 Chicago, IL, USA

Zhejiang University June 2025
B.S. in Physics, School of Physics — GPA: 3.88/4.0 Hangzhou, Zhejiang, China
Thesis advisor: Prof. Renyue Cen.

RESEARCH INTERESTS

I am keen on the physics of structure formation and evolution across physical scales, from cosmological structures to galaxy formation and turbulence. I have great interest in probing physical processes by building and analyzing numerical simulations, as well as testing them against observations. I am also interested in connecting scales, e.g., optimizing priors for large-scale models using small-scale physics.

RESEARCH EXPERIENCE

Turbulence Structures in the Multiphase ISM Aug. 2026 — present
Visiting Student, Princeton University; advised by Dr. Chang-Goo Kim and Dr. Rajsekhar Mohapatra

- Decompose turbulence in the TIGRESS simulations into compressive and solenoidal modes with independent methods.
- Mask the hot gas before decomposition to isolate the turbulent modes of the warm and cold phases, and relate them to feedback and star formation.

Thin Disks as a Function of Baryonic Mass Dec. 2025 — present
Graduate Researcher, advised by Prof. Andrey Kravtsov

- Develop a pipeline to process H I-selected dwarf galaxies in deep imaging with cleaning, deblending, segmentation, and masking.
- Measure the morphology of the selected galaxies with different methods.
- Model the dependence of disk thickness on stellar mass, baryonic mass, and rotation velocity.

Planar Structures in ΛCDM Universes Nov. 2023 — Sept. 2026
Undergraduate/Graduate Researcher, advised by Prof. Renyue Cen, in collaboration with Prof. P. J. E. Peebles

- Developed a geometric probe for planar structures and optimized a computationally intensive pipeline for Local Group Analogs (LGAs).
- Used LGAs in cosmological N-body simulations to assess how rare the observed local planar structures are in ΛCDM and analyzed the statistical properties of planar structures.

The Quasar Dipole in Cosmological Simulations Winter 2026
Independent course project, Statistical Methods in Astrophysics

- Built mock quasar skies for Local Group-like observers in large N-body simulations.
- Tested how unusual the observed quasar dipole and its misalignment with the CMB dipole are in ΛCDM.

PUBLICATIONS

– **Jiang, X.**, Cen, R., & Peebles, P. J. E. "Planar structures in ΛCDM universes." *in prep.*
– **Jiang, X.** & Kravtsov, A. "The mass threshold for thin disks." *in prep.*

TEACHING

Graduate Student Teaching Assistant — BIOS 21403: Quantitative Bioimaging Spring 2026
The University of Chicago, Biological Sciences Collegiate Division

- Led discussion sections, held office hours, and graded for a course on image analysis in Python.

CONFERENCES & PRESENTATIONS

Open SkAI 2025 — attendee Sept. 2025
27th Guo Shoujing Symposium on Galactic Cosmology — poster presenter June 2025

HONORS & AWARDS

Outstanding Student, Zhejiang University 2025
Excellent Graduate, Zhejiang University 2025
First Class Scholarship, Zhejiang University 2022

COURSEWORK

* graduate course

Mathematics
Approximation Algorithms*; Foundations of Machine Learning and AI*; Mathematical Analysis; Advanced Algebra and Analytic Geometry; Ordinary Differential Equations; Methods of Mathematical Physics; Probability and Statistics

Physics
Cosmology I*; Galaxies*; Statistical Methods in Astrophysics*; Quantum Mechanics; Electrodynamics; Theoretical Mechanics; Statistical Mechanics; Computational Physics; Galactic Astronomy

SKILLS

Programming: Python, R, C++, SQL, LaTeX, Bash
Computational: HPC (multi-core CPU and GPU), Slurm
Methods: Machine Learning, Monte Carlo, Bayesian Inference, MCMC, Fourier Analysis
Languages: Mandarin (native), English (fluent)